⛈️ The Strike & Staying Safe: Guided Notes

Chemistry · Unit 1C · Level 3 (closer to the class work) · goes with L1.10-L1.14 (Energy Flow, The Strike, Conductors & Insulators, No Swimming, Unit Test)

Name: _____

Use the class slides and your reading. Key words:

potential energy | *thermal energy* | *stepped leader* | *return stroke* | *conductor* | *insulator* | *ion* | *current* | *dissolve* | *lightning rod*

A. Energy flow (L1.10)

1. Separated charge stores _____ energy.

2. List the three forms of energy that potential energy changes into during a strike.

_____

B. The strike (L1.11)

3. Fill in the table.

Step	What happens
1. Stepped leader	
2. Streamer	
3. Return stroke	
4. Thunder	

C. Conductors and insulators (L1.12, L1.13.5)

4. In a _____, electrons move easily. In an _____, they do not.

5. Why are metals good conductors? Use the word electrons.

_____

_____

6. Fill in the table.

Material	Conductor or insulator?
copper wire	
rubber	
salt water	
dry wood	

D. Salt water and safety (L1.13, L1.14)

7. **Reading the No Swimming data: 1 amp (A) = 6.28×10^{18} charges per second.**

> Example: 2 A = $2 \times 6.28 \times 10^{18} = 12.56 \times 10^{18} = 1.256 \times 10^{19}$ charges per second.

a) 0.5 A = ____ $\times 10^{18}$ charges per second _____

b) 3 A = ____ $\times 10^{18}$ charges per second _____

8. Claim: How does dissolving salt in water affect the current? Give evidence from the lab and reasoning.

9. Draw a house with a lightning rod. Use arrows to show the path of the charge.